

COURSE DESCRIPTION

Course Number	0418 - 470	Course Title	Database Systems
Credit Hours	4 (3:3:4)	Course Coordinator	Dr. Mohamed Almulla
Compulsory	Yes	Prerequisite	0418-301

Textbook

Title:	Fundamentals of Database Systems.
Author:	R. Elmasri, S. Navathe.
Publisher:	Addison Wesley Publishing Company, 2010.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: Information models and systems. The relational data model, database architecture, database management systems.	4
2	Data Modeling, Security, Privacy, Protection.	8
3	Structured Query Language (SQL).	10
4	Procedural Language (PL/SQL): Block structures, cursors (implicit and explicit), triggers, and exception handling.	8
5	Database Design: Data collection, conceptual design, relationship and participation constraints, mapping ER-diagrams, design anomalies. Design methods and algorithms based on functional-dependencies, normal forms.	6
6	Transaction Processing: Transactions in multi-user databases, concurrency, locking protocols, recovery.	3
7	Exams.	3
	Total.	42

Laboratory Work

	Topics	weeks
1	Integrated Oracle Environment.	1
2	Creating user, changing password, user's privileges.	2
3	Creating database schema, tables, constraints, primary and foreign keys.	3
4	Populate database with data, validation of input, different formats of data.	4
5	Basic queries against the database, single table.	5
6	Composed queries with join of tables, and other set operations on relations.	6
7	Using aggregate functions in queries.	7
8	PL/SQL basics, structure and blocks.	8
9	Procedures and functions in PL/SQL.	9
10	Creating and application of triggers.	10
11	PL/SQL Packages- organization and applications.	11
12	Multi-user environment, transactions, snapshots.	12
13	Backup and Recovery procedures in database.	13

Grading

15-20%	Midterm 1
15-20%	Midterm 2
15-20%	Lab
10-5%	Hmk/quiz
10-5%	Project
40%	Final

Course Outcome

Students should be able to:

1. Understand the basic database concepts and relational theory.
2. Understand and develop relational / conceptual data model.
3. Use and apply data definition and data manipulation languages like SQL.
4. Apply functional dependencies and normalization concepts in database modeling and design.
5. Understand the basics of database transactions, concurrency controls and database recovery.
6. Do research about new topics/trends in databases.
7. Design, develop and present a workable DBMS project as part of a team to solve real-life problem.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M					
2	H	L				M
3	H					
4	M					M
5	L					
6	M	H	H		M	M